

Practical No. 01

Connecting Two Devices Using Cisco Packet Tracer, Easiest Revision Notes

Terms You Should Know First

Read these once before the steps, everything below will feel much easier.

Cisco Packet Tracer

A free simulator made by Cisco that lets you build and test computer networks on your laptop without buying real hardware.

End Devices

Devices that generate or receive data, like PCs, laptops, printers or servers. In this practical the end devices are our two PCs.

Cross Over Cable

A cable used to connect two similar devices directly, like PC to PC or switch to switch. Its wiring is arranged so the send and receive pins line up correctly between two similar devices.

Straight Through Cable

A cable used to connect two different types of devices, like a PC to a switch. It is the more common cable type.

FastEthernet Port

The network port on a PC or switch where the network cable physically plugs in. It supports data speeds up to 100 Mbps.

IP Address

A unique numerical label given to a device on a network, such as 192.168.1.1, so that other devices can find and reach it.

Subnet Mask

A number like 255.255.255.0 that tells a device which part of an IP address represents the network and which part represents the specific device.

PDU

Protocol Data Unit. It simply means a unit of data being sent across the network. Packet Tracer uses this term when you send a test message between devices.

Add Simple PDU

A quick testing tool in Packet Tracer that sends one test packet from a source device to a destination device so you can check if the connection actually works.

What You Actually Did

- Opened Cisco Packet Tracer and picked two PCs from End Devices in the bottom toolbar.
- Selected the Cross Over cable from Connections and joined the two PCs through their FastEthernet0 ports.
- Opened each PC, went to Desktop, then IP Configuration, and gave each PC an IP address.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

After giving both PCs an IP address, you tested the connection using Add Simple PDU. You clicked once on PC0 and once on PC1, and the PDU list at the bottom showed the word Successful, which

meant the two computers could talk to each other properly.

The Simple Way To Remember This Whole Practical

Three things happen in order. First you make the physical connection with a cable. Second you give both devices an address so they know who they are on the network. Third you test it by sending one small message and checking if it says Successful. That is really the whole practical in three lines.

Step Map, So You Never Forget The Order

Step	What You Do	Why
1	Place two PCs on the workspace	You need two end devices to connect
2	Pick Cross Over cable and connect FastEthernet0 on PC0 to FastEthernet0 on PC1	Two FastEthernet devices need a cross over cable to talk directly
3	Open Desktop, then IP Configuration on each PC	Every device needs its own IP address to be identified
4	Assign 192.168.1.1 to PC0 and 192.168.1.2 to PC1	Both PCs must be in the same network range to communicate
5	Use Add Simple PDU and click PC0 then PC1	PC0 sends one test message to check the connection
6	Check the PDU list for the word Successful	This confirms the network is actually working

Exercise Given By The Teacher

Naming rule for devices: RollNumber_DeviceName

Connect 2 PCs directly and show the connection working.

PC1 IP address should be 192.168.x.1

PC2 IP address should be 192.168.x.2

Here x is your own roll number.

Viva Questions

What is Cisco Packet Tracer used for?

It lets you design, build and test a computer network virtually before or instead of using real hardware.

Why did you use a cross over cable instead of a straight cable?

Because we were connecting two similar devices directly, a PC to a PC. Cross over cables are made for connecting similar devices, straight cables are for connecting different devices like a PC to a switch.

What is the difference between straight through and cross over cable?

A straight through cable connects two different device types, like PC to switch. A cross over cable connects two similar device types, like PC to PC or switch to switch.

What is an IP address?

A unique number assigned to a device so it can be identified and reached on a network.

What is a subnet mask and why do we set it?

It tells the device which part of an IP address is the network part and which part is the device part. Both PCs need matching subnet masks to be considered part of the same network.

What does PDU mean?

Protocol Data Unit, a general term for a unit of data moving through the network.

How do you know two devices are connected properly?

By sending a simple PDU between them and checking that the result shows Successful in the PDU list.

At which OSI layer does a plain network cable work?

Physical Layer, Layer 1, since it only deals with the physical transmission of signals.

Why must PC0 and PC1 be in the same IP range?

Devices can only communicate directly on a LAN if they belong to the same network range, otherwise a router would be required.

Important Points For Interviews And Future Practicals

- Cross over cables are becoming less important in real life because modern network cards support Auto MDI X, meaning they can automatically detect and adjust for either cable type. Interviewers still ask the concept though, so know it well.
- The real world equivalent of this practical is setting up any small home or office network, the same idea of physical connection plus addressing plus testing applies everywhere.
- Ping and Add Simple PDU do the same basic job, testing connectivity, but Add Simple PDU is a Packet Tracer specific tool while ping is a real command used in actual networks.
- Two devices being in the same subnet is one of the most common troubleshooting checks in real networking jobs, so this basic exercise is actually testing a core skill.
- This practical is Layer 1 and Layer 3 territory, physical cabling is Layer 1 and IP addressing is Layer 3. Practical 2 builds on this by introducing Layer 2 concepts through ARP, and Practical 4 adds routing between two separate networks.

Revise in this order, terms first, then what you did, then the step map, then the viva questions.